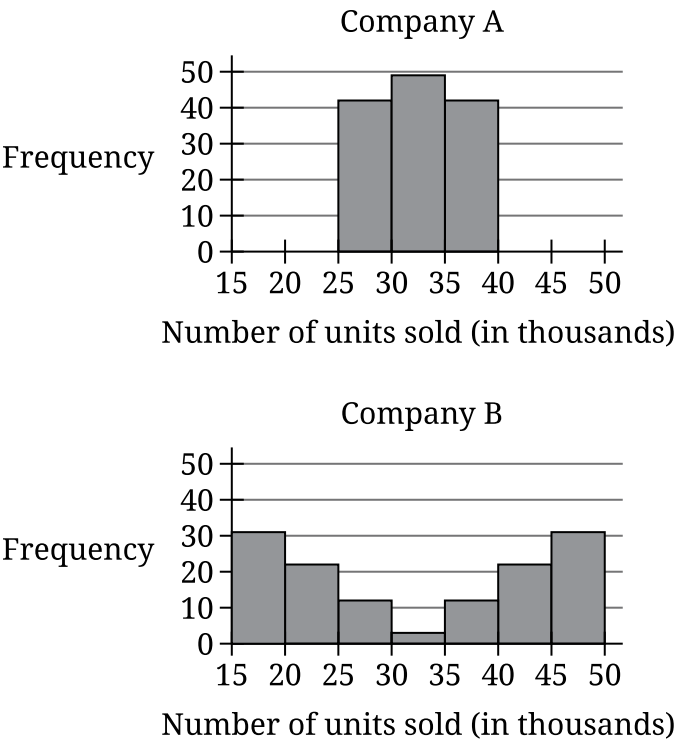


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Medium

Question



The histograms summarize the distributions of number of units sold, in thousands, for company A and company B. Which statement best compares the standard deviations of number of units sold for these companies?

- Answer**
- A. The standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.
 - B. The standard deviation of number of units sold for company A is greater than the standard deviation of number of units sold for company B.
 - C. The standard deviation of number of units sold for company A is equal to the standard deviation of number of units sold for company B.
 - D. There is not enough information to compare the standard deviations.

Correct Answer: A

Rationale

Choice A is correct. Standard deviation measures the spread of a given data set from its mean. In a data set with a smaller standard deviation, there are more values close to the mean. In a data set with a greater standard deviation, there are more values farther from the mean. The two histograms shown have the same scale on the horizontal axis. Therefore, their standard deviations can be compared by visually comparing the spreads of their histograms. The distribution summarized by each histogram is symmetric. Therefore, the mean of the data set for each histogram is a value in the middle bar of that histogram. The middle bar of each histogram has a value of at least **30** thousand units sold but less than **35** thousand units sold. Therefore, the mean of the data set for each histogram is at least **30** thousand and less than **35** thousand. The histogram for company A shows all the values in that data set are close to the mean. For company B, the histogram shows there are fewer values

close to the mean and more values farther from the mean. Therefore, the standard deviation of number of units sold for company A is less than the standard deviation of number of units sold for company B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.